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DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9744/02

Paper 2 Structured Questions

14 September 2018

2 hours

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

Answer **all** questions.

Write your answers on space provided in the Question Paper.

INFORMATION FOR CANDIDATES

Essential working must be shown.

The intended marks for questions or parts of questions are given in brackets [].

For Examiner's Use	
1	/ 10
2	/ 12
3	/ 10
4	/ 7
5	/ 9
6	/ 12
7	/ 14
8	/ 8
9	/ 8
10	/ 10
Total [100]	

This document consists of **28** printed pages (including **2** blank pages).

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Structured Questions (100 marks)
Answer **all** questions in this section.

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Question 1

(a) Fig. 1.1 below is an electron micrograph showing part of an animal cell.

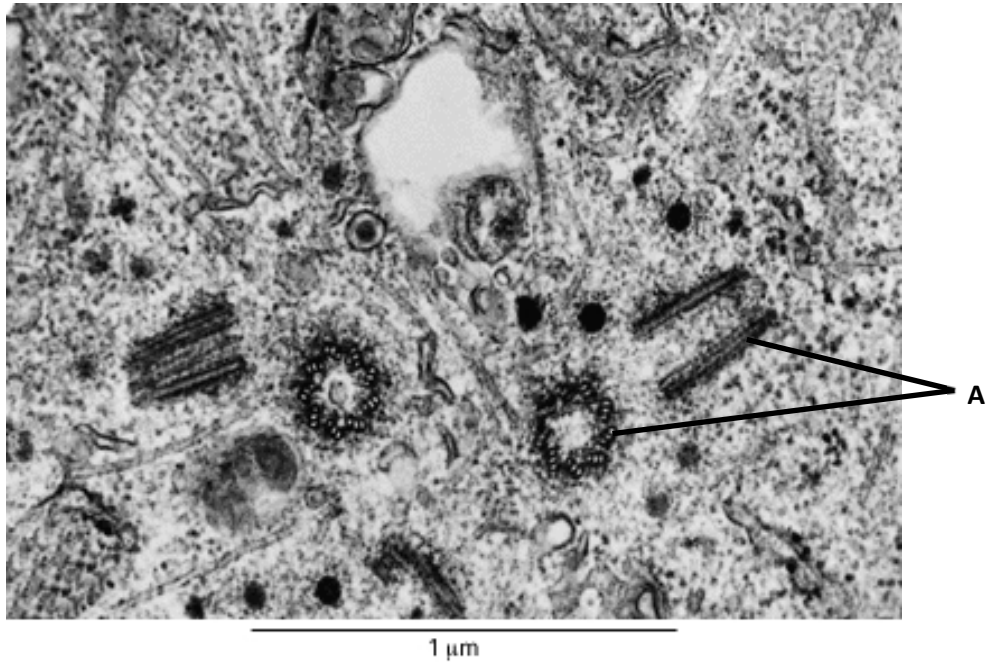


Fig. 1.1

(i) Name the structures labeled A. [1]

A:

(ii) Name the process the cell is undergoing and explain your answer. [2]

- (b)** Scientists often employ the method of fluorescence imaging to study the process of nuclear division in cells. This method involves the use of fluorescence tagged antibodies to label specific structures within the cell, so that their movements and behaviours can be followed over the course of nuclear division.

A group of scientists decided to employ the aforementioned method to study the changes in centromere number in an animal cell undergoing meiosis. Initial testing showed that each centromere could be visualised as a single dot and therefore the total number of dots can be taken to correspond to the total number of centromeres within the cell. Table 1 shows the number of dots and hence centromeres observed per cell, at various stages of meiosis.

Table 1

Stage of Meiosis	Prophase I	Prophase II	Anaphase II	Telophase II
Number of dots observed per cell	16			16

With reference to Table 1,

- (i)** Predict the number of dots (i.e. centromeres) observed in prophase II and anaphase II respectively by filling out Table 1. [1]
- (ii)** Explain your answer to part **(b) (i)**. [2]

When the same experiment was conducted while the cell was in interphase, prior to meiosis I, no distinct dots could be observed in the nucleus. Instead, the entire nucleus emitted fluorescence.

- (iii)** Suggest a reason for this observation. [1]

Apart from using fluorescence imaging to determine the changes in centromere numbers, this technique was also employed by the scientists to study the movement of chromosomes during meiosis in the animal cell. More specifically, the changes in distance between 2 corresponding regions of non-sister chromatids of a pair of homologous chromosomes (**X**) were measured over the course of the first meiotic division. The results are shown in Fig. 1.2 below.

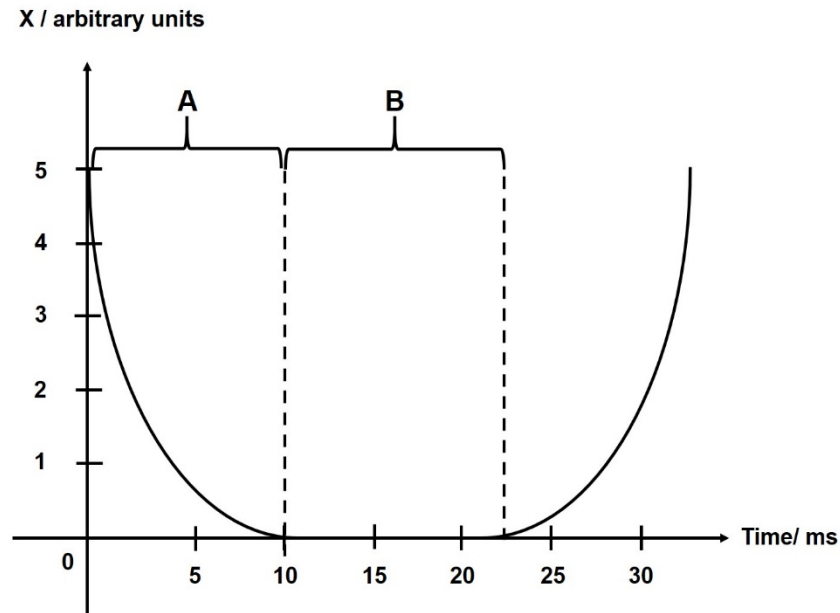


Fig. 1.2

With reference to Fig. 1.2, explain the changes in **X** for

(iv) Region A. [1]

(v) Region B. [1]

Driven by the success of the above experiment, one of the scientists decided to conduct the same experiment for the second meiotic division of the animal cell. His investigations however did not produce any results.

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(vi) State a reason for the lack of results obtained. [1]

Total: [10]

Question 2

A study was conducted to study the inheritance of coat colour in mice, in which one of the allele is also known to affect normal embryonic development (lethal).

A cross between agouti mouse and yellow mouse resulted in half of the F_1 progeny being agouti mice and the other half being yellow. Mating of F_1 yellow mice resulted in the following F_2 generation.

Agouti mice	98
Yellow mice	202

- (a) Using the symbols **A** and **a** for the two alleles involved, draw a genetic diagram in the space below to show how the F_1 cross resulted in the F_2 progeny. [3]

- (b) In another experiment involving deer mouse, pure breeding pink-eyed mice with wild-type fur was crossed with pure breeding dark-eyed albino mice.

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The resulting progeny all had wild-type fur and dark eyes. These F₁ mice were then crossed with pink-eyed albino mice.

The results are shown in Table 2. It was difficult to distinguish between mice that are dark-eyed albino and pink-eyed albino, so these two phenotypes were counted together (F.H. Clark. 1936. Journal of Hereditary 27:259-260).

Table 2

Phenotype	Number of progeny
Wild-type fur, dark-eyed (recombinant)	12
Wild-type fur, pink-eyed (parental)	62
Albino, dark-eyed (parental)	78
Albino, pink-eyed (recombinant)	

A chi-square test was conducted to determine if the genes for fur type and eye colour assort independently.

- (i) Complete the following table to find the calculated chi-square value. [2]

Phenotype	Observed (O)	Expected (E)	(O-E) ² / E
Wild-type fur, dark-eyed	12		
Wild-type fur, pink-eyed	62		
Albino, dark-eyed Albino, pink-eyed	78		
Chi Square Value			

The chi square table is provided below:

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(ii) What conclusion can be drawn from the chi-square test in (b)(i)? [3]

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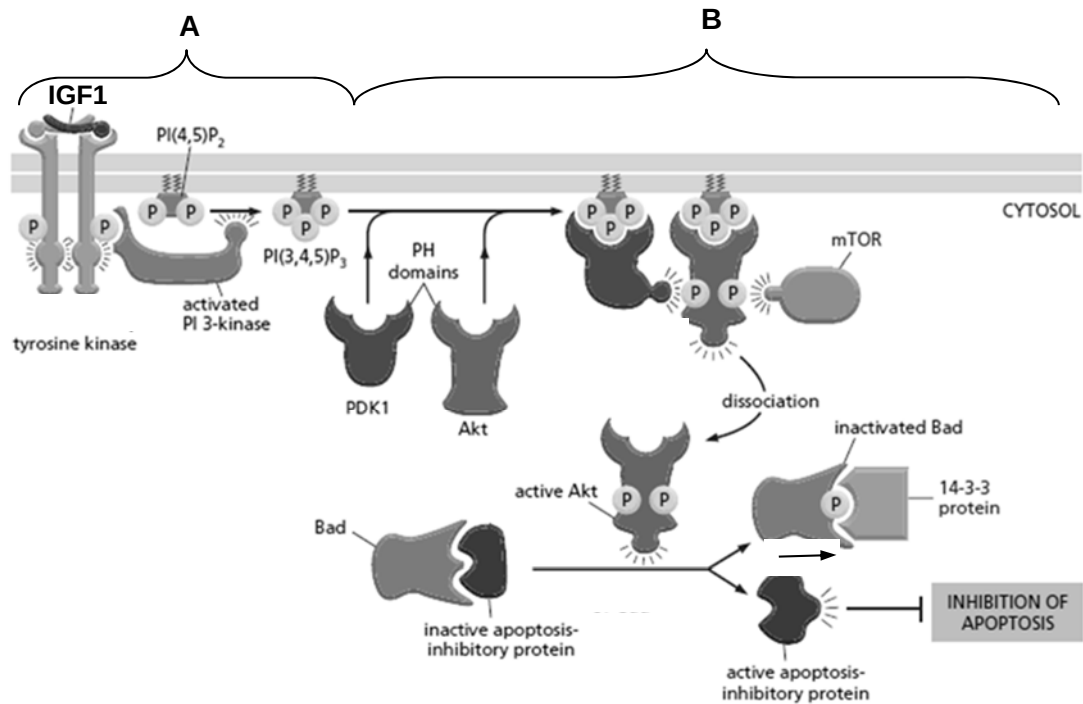
(c) Using **E/e** to represent alleles for eye colour and **A/a** to represent alleles for fur coat colour, explain the result of the F_1 cross in the deer mouse experiment using a genetic diagram. [4]

Total: [12]

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Question 3

Many types of animal cells need to receive chemical signals to survive and grow. Fig. 3 shows how a cell receives Insulin-like Growth Factor 1 (IGF1) and prevents the cell from triggering apoptosis (cell death).

**Fig. 3**

(a) Describe stages A and B shown in Fig. 3. [6]

A:

B:

Scientists have found that the PI 3-kinase and Bad protein of the IGF1 signalling pathway has been mutated in tumour cells. Tumour cells are uncontrolled dividing cells which are resistant to apoptosis.

(b) Describe a possible mutation of the following proteins and the corresponding effect of the mutation of the protein's activity in tumour cells.

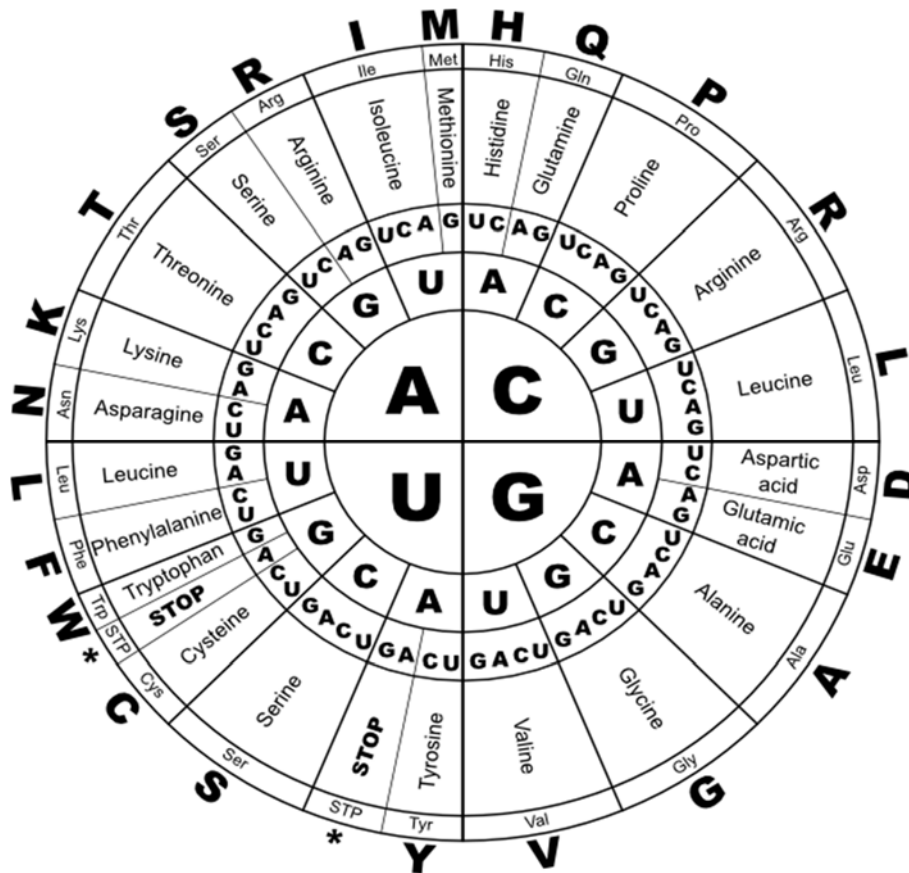
(i) PI 3-kinase [2]

(ii) Bad protein [2]

Total: [10]

Question 4

The *APP* gene provides instructions for making a protein called amyloid precursor protein. This protein is found in many tissues and organs, including the brain and spinal cord. The most common mutation on the *APP* gene involves one codon being changed from GCC to GUC. Fig. 4.1 shows the genetic code.



(b) With reference to Fig. 4.1, describe the mutation that has occurred. [2]

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(c) Amyloid precursor protein is cut by enzymes to create smaller fragments, some of which are released outside the cell. Two of these fragments are called soluble amyloid precursor protein (sAPP) and amyloid beta (β) peptide. This mutation in the *APP* gene can lead to the production of a “stickier” form of the β peptide. When these protein fragments are released from the cell, they can accumulate in the brain and form clumps called amyloid plaques. These plaques are characteristic of Alzheimer disease.

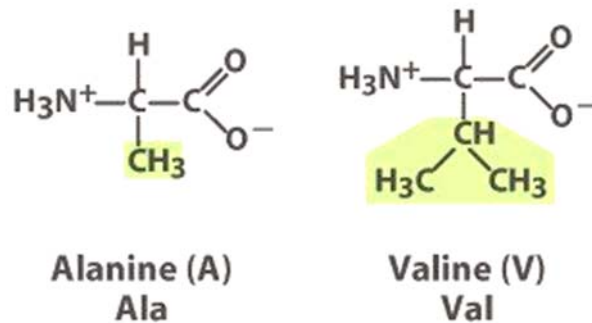


Fig. 4.2

Fig. 4.2 shows the structures of alanine and valine. Both amino acids contains non-polar R group. Suggest how a change from alanine to valine can result in a mutated amyloid precursor protein which then give rise to amyloid plaques. [2]

Total:[7]

Question 5

Fig. 5 shows the double helix structure of DNA discovered by Watson & Crick in 1953.

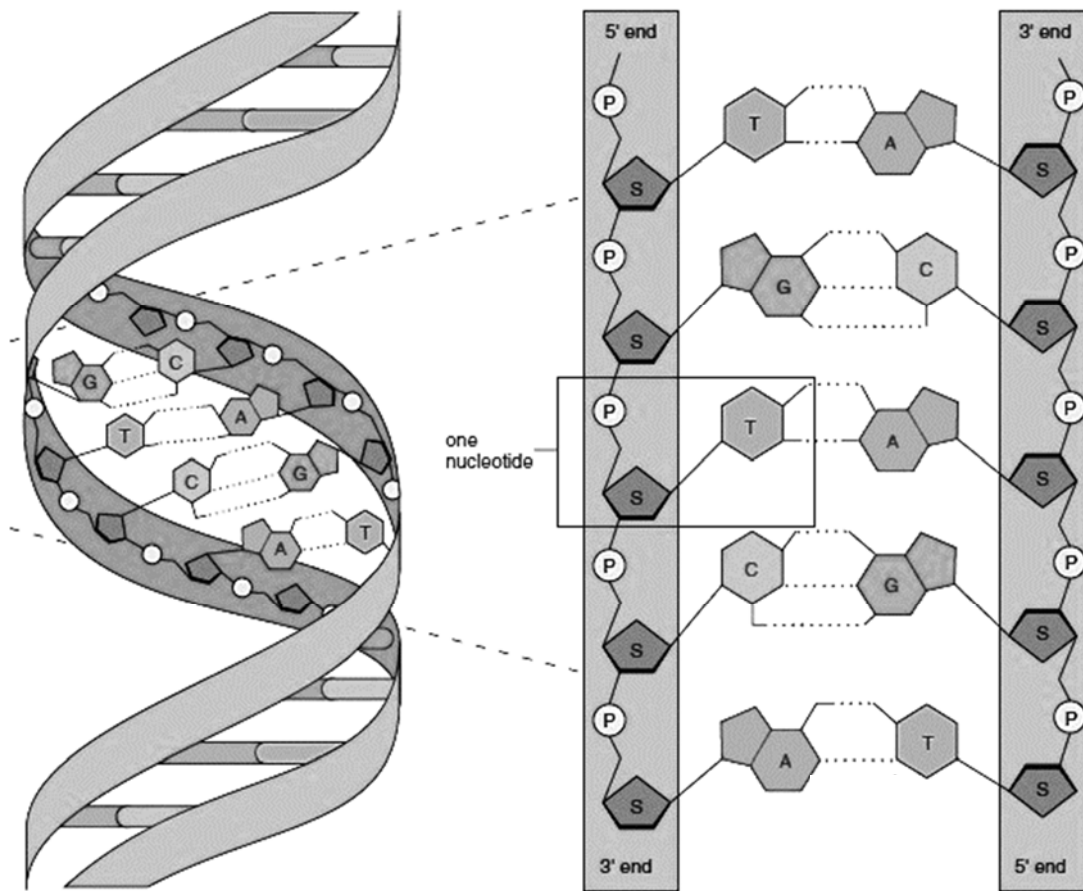


Fig. 5

(a) Explain how the stability of the DNA structure is maintained. [3]

- (b)** RNA molecules that possess catalytic activity are known as ribozymes. One well-known ribozyme is found in the ribosome. The catalytic site of ribosomes, which holds the growing peptide chain and aminoacyl-tRNA in place, is made of RNA.

- (i)** State the name of the enzyme that is found in ribosomes and describe the reaction it catalyses. [2]

- (ii)** Suggest how the RNA molecules form the catalytic site of the ribosome. [3]

- (c)** Lactimidomycin, a potential anti-cancer drug, is found to selectively inhibit translation. Laboratory studies involving lactimidomycin have resulted in an accumulation of dipeptides in the cells.

Suggest how lactimidomycin works to inhibit translation. [1]

Total:[9]

Question 6

Dengue is the most rapidly spreading mosquito-borne viral disease in the world. In the last 50 years, incidence has increased 30-fold with increasing geographic expansion to new countries and, in the present decade, from urban to rural settings. An estimated 50 million dengue infections occur annually and approximately 2.5 billion people live in dengue endemic countries.

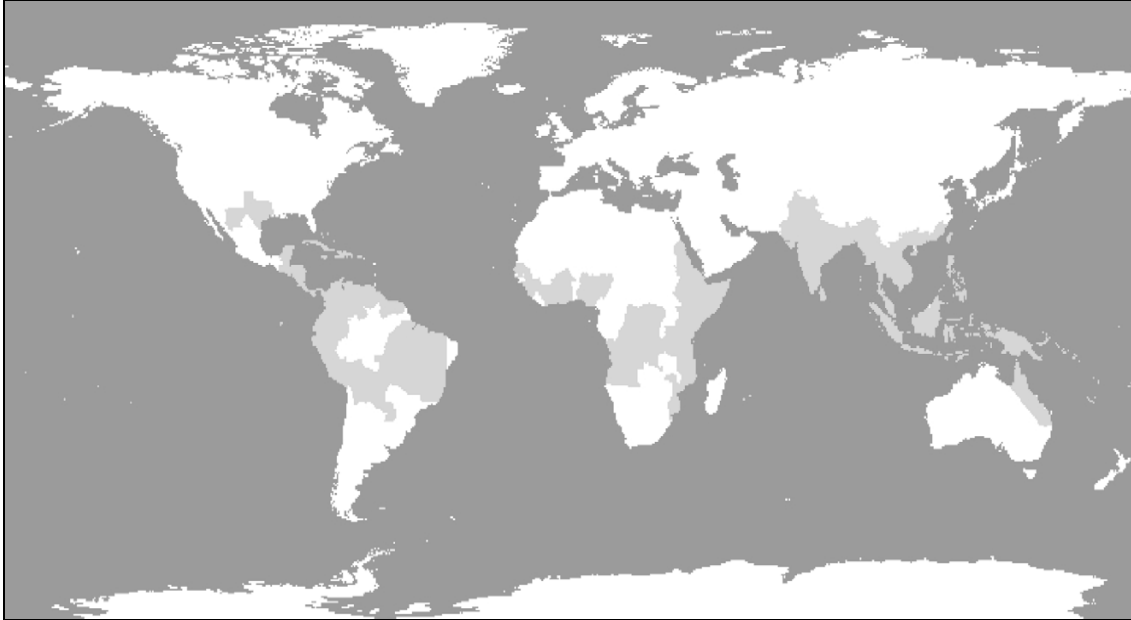


Fig. 6

Shaded areas in Fig. 6 represents countries at risk of dengue fever due to presence of *Aedes* mosquito, as of 2008.

(a) Outline the developmental stages in the life cycle of the *Aedes* mosquito. [4]

- (b) Explain why the growing area of influence of dengue fever overlaps with that of the *Aedes* mosquito. [3]

- (c) To some extent the range of the *Aedes* mosquito has also followed human expansion. Explain how this may be true. [2]

- (d) An entomologist reviewed ideas about the history of *A. aegypti*'s distribution throughout the world. It is almost certain that the ancestor of the domestic form of *A. aegypti* lived in sub-Saharan Africa.

The larval habitat was likely tree holes and non-human animals provided blood meals.

Today, this ancestral form still exists in forests and vegetated ecotones in sub-Saharan Africa and is called by a subspecies name, *formosus*.

Using molecular techniques, describe how scientist can determine if *formosus* and current day of *A. aegypti* are from the same species. [3]

Total:[12]

Question 7

Recent experiments have demonstrated that the anti-malarial drug, chloroquine (CQ), has inhibitory effect against the *in vitro* replication of human influenza A virus, H1N1.

Chloroquine is a weak base which interferes with pH-dependent steps of the reproductive cycle of influenza viruses. Fig. 7.1 shows the results of an experiment performed where cells were incubated with CQ after viral inoculation and tested for cell viability (survival) and viral RNA copies at 24-hour post-infection.

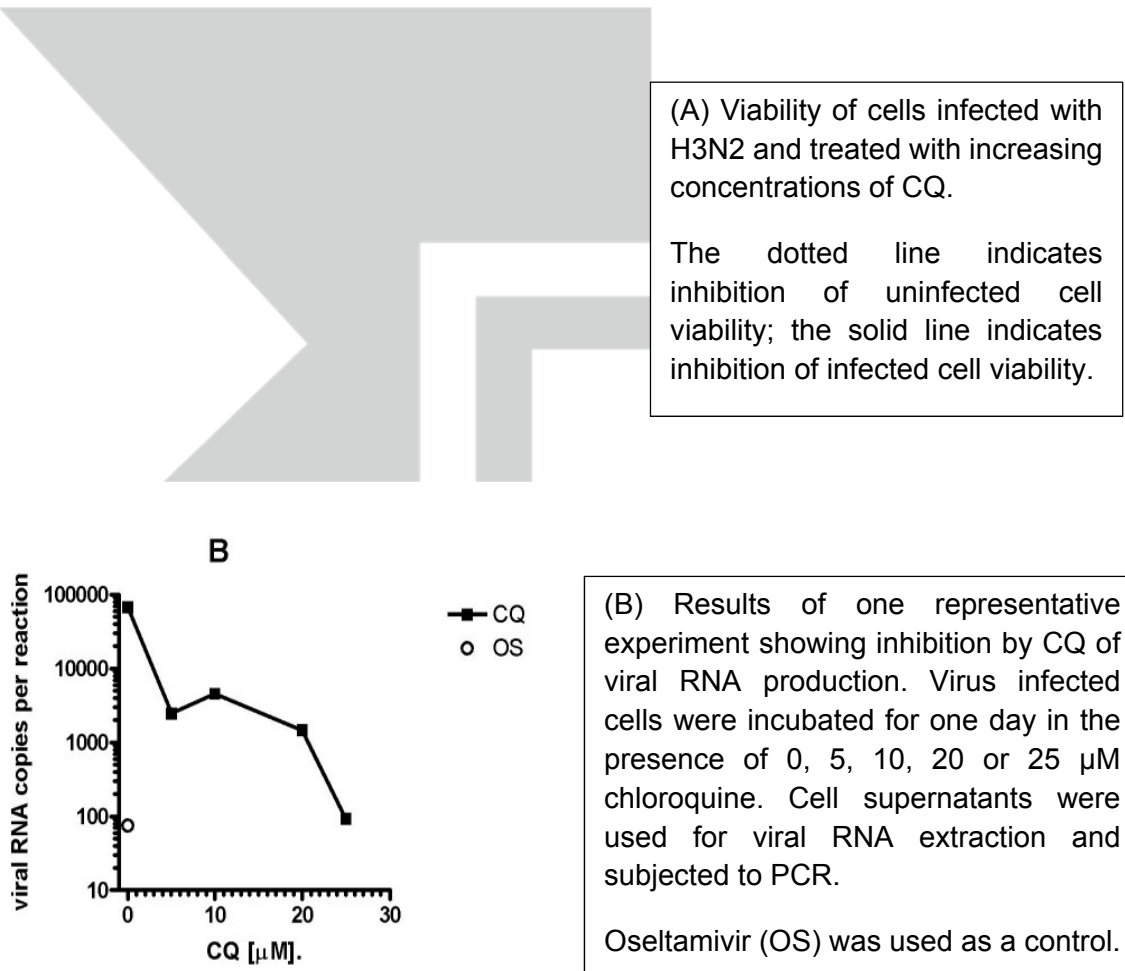


Fig. 7.1

- (a) (i) With reference to Fig. 7.1, describe the effect of chloroquine on infected cells *in vitro*. [3]

- (ii) From the information provided, suggest how chloroquine inhibits influenza viral replication. [1]

- (b) Outline how the influenza virus is able to bypass the human defense mechanism to cause disease. [4]

- (c) State the main differences between the structure and organisation of prokaryotic and eukaryotic genomes. [6]

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Total: [14]

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Question 8

Most cases of cervical cancer are caused by infection with Human Papilloma Virus (HPV). This virus can be spread by sexual contact. There are many types of HPV, each identified by a number. Most of these types are harmless but types 16 and 18 are most likely to cause cervical cancer.

A vaccine made from HPV types 16 and 18 is offered to girls aged 12 to 13. Three injections of the vaccine are given over six months. In clinical trials, the vaccine has proved very effective in protecting against HPV types 16 and 18. However, it will be many years before it can be shown that this vaccination programme has reduced cases of cervical cancer. Until then, smear tests will continue to be offered to women, even if they have been vaccinated. A smear test allows abnormal cells in the cervix to be identified so that they can be removed before cervical cancer develops.

The Department of Health has estimated that 80% of girls aged 12 to 13 need to be vaccinated to achieve herd immunity to HPV types 16 and 18. Herd immunity is where enough people have been vaccinated to reduce significantly the spread of HPV through the population.

Use information from this passage and your own knowledge to answer the following questions.

- (a) HPV vaccine is offered to girls aged 12 to 13 (line 5). Suggest why the vaccine is offered to this age group. [1]

- (b) The vaccine is made from HPV types 16 and 18 (line 5). Explain why this vaccine may **not** protect against other types of this virus. [2]

- (c) It will be many years before it can be shown that this vaccination programme has reduced cases of cervical cancer (lines 7 to 9). Suggest **two** reasons why. [2]

- (d) Smear tests will continue to be offered to women, even if they have been vaccinated (lines 9 to 10). Suggest why women who have been vaccinated still need to be offered smear tests. [1]

- (e) Some doctors suggested offering the vaccine to young men. Explain why it would be advantageous to vaccinate young men along with young women. [2]

Total: [8]

Question 9

Cranes are large birds. One of the earliest methods of classifying cranes was based on the calls they make during the breeding season.

- (a) Explain why biologists could use calls to investigate relationships between different species of crane. [2]

More recently, biologists have used DNA hybridisation to confirm the relationships between different species of crane. They made samples of hybrid DNA from the same and from different species. They measured the percentage of hybridisation of each sample. The results are shown in Table 9.

Table 9

Species of crane from which hybrid DNA was made	Percentage DNA hybridisation / %
<i>Grus americana</i> and <i>Grus monachus</i>	97.4
<i>Grus monachus</i> and <i>Grus rubicunda</i>	95.7
<i>Grus americana</i> and <i>Grus rubicunda</i>	95.5
<i>Grus rubicunda</i> and <i>Grus rubicunda</i>	99.9
<i>Grus americana</i> and <i>Grus americana</i>	99.9
<i>Grus monachus</i> and <i>Grus monachus</i>	99.8

- (b) The biologists measured the temperatures at which the samples of hybrid DNA separated into single strands. Explain why these temperatures could be used to find the percentage of DNA hybridisation. [2]

The bone protein, osteocalcin, has been extracted from fossil Neanderthal skulls found in Shanidar Cave, Iraq. The skulls were estimated to be approximately 75 000 years old.

Using mass spectrometry, the amino acids of this Neanderthal osteocalcin have been sequenced. This sequence has been compared with that in osteocalcin from *Homo sapiens* and other modern primates.

The sequences of the first 20 amino acids in the primary structure of osteocalcin from this study are shown below in Fig. 9. Amino acids are represented by capital letters.

	1	10	20
Modern human	Y	L	Y
Neanderthal	Y	L	Y
Chimpanzee	Y	L	Y
Orang-utan	Y	L	Y
Gorilla	Y	L	Y
Monkey	Y	L	Y

Fig. 9

- (c) Suggest what the data indicate about the relationships between Neanderthals and modern primates. Give reasons for your answer. [4]

Total: [8]

Question 10

The following experiment was carried out to investigate the effect of light intensity on the rate of photosynthesis of a water plant, *Elodea*.

- *Elodea* was cut into three pieces, each 10 cm long.
- Each piece of *Elodea* was placed in a glass tube, containing 0.5% sodium hydrogen carbonate solution, which was then sealed with a bung.
- Tube **A** was placed 10 cm away from a lamp.
- Tube **B** was placed 5 cm away from a lamp.
- Tube **C** was placed in a dark room.
- An oxygen sensor was used to measure the percentage of oxygen in the solutions at the start of the experiment and again at 5, 10 and 20 minutes.

The results are shown in Fig. 10.1.

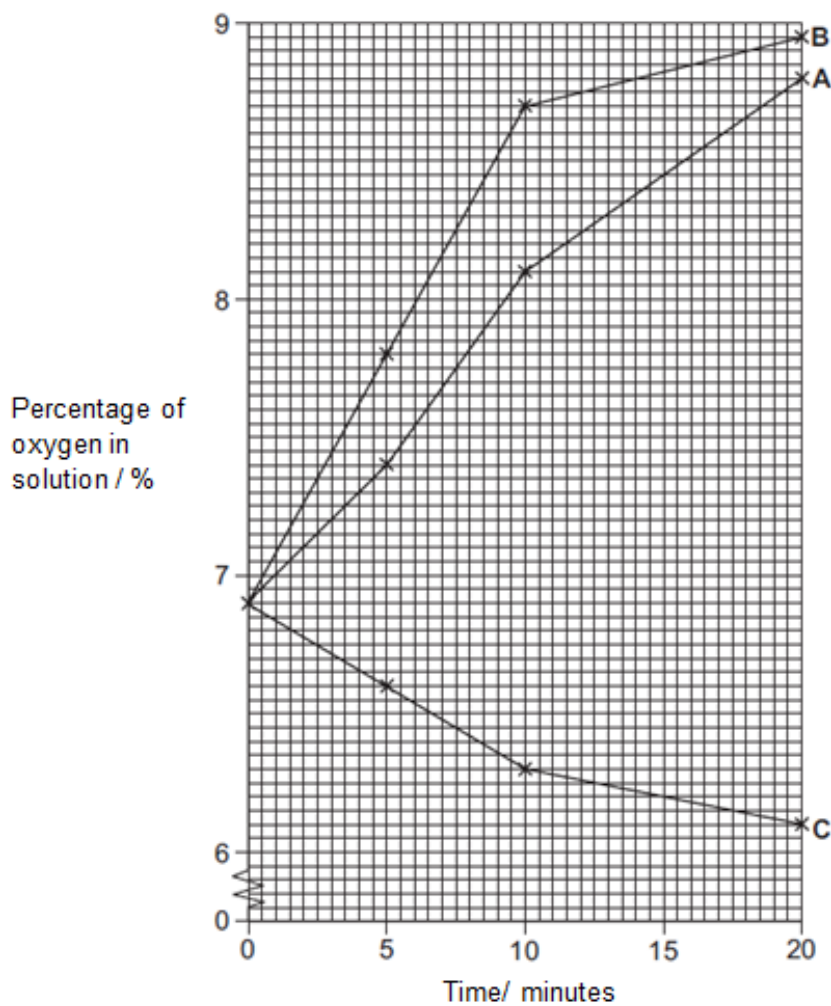


Fig. 10.1

(a) With reference to Fig. 10.1,

(i) Calculate the rate of oxygen production for tube **A** for the 20 minutes of the experiment. Give your answer to two decimal places. Show your working. [1]

(ii) Compare the results for tubes **A** and **B**. [2]

(iii) Explain the results for tube **C**. [2]

Fig. 10.2 outlines the early stages of respiration.

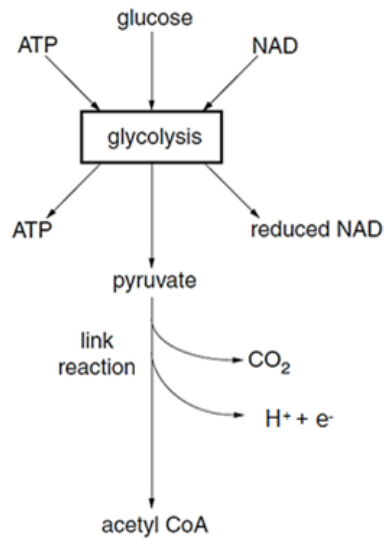


Fig. 10.2

(b) With reference to Fig.10.2,

(i) Explain why ATP is needed at the start of glycolysis. [1]

(ii) Explain the fate of the hydrogen atoms released by pyruvate during the link reaction. [4]

Total: [10]

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